

MCRP: A Prospective, Claim-Versioned Verification Lifecycle for Computational Research

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Abstract

Computational research is often published as a narrative article accompanied by a repository, an environment description, and selected data or outputs. These materials can support inspection and rerunning, but they do not by themselves identify which execution supports each material claim, what transformations intervene between source data and reported values, which dependencies remain trusted rather than re-examined, or which human judgments were required for acceptance. Agentic tools make this distinction more consequential: an agent can reconstruct environments, execute workflows, compare outputs, and assemble provenance, yet a successful automated run cannot establish scientific validity.

We propose the Minimum Credible Reproducibility Protocol (MCRP), a prospective protocol for constructing and reviewing an immutable Scientific Record Release. MCRP treats material claims, evidence, outputs, executions, workflow definitions, configurations, environments, source artifacts, and declared trust dependencies as versioned review objects. Verification is externally enforced: author-controlled execution may contribute evidence, but conformance requires a verifier operating under a separately recorded authority, and scientific acceptance remains bound to qualified human judgment. Any material change creates a new release identity and requires affected checks and decisions to be re-established. MCRP also separates assurance from resource burden through a multidimensional resource profile and distinguishes full, slice, checkpoint, and witness assessment modes.

This technical supplement defines the lifecycle and its limits, relates it to established reproducibility, provenance, archival, and review practices, and describes a small synthetic reference implementation. The current implementation has passed schema, replay, and adversarial fixture checks, but it has not received independent scientific sign-off, been evaluated on a real scientific publication, or demonstrated improved review outcomes. We therefore present MCRP as a testable protocol proposal rather than a certification standard or validated intervention.

1. Introduction

Scientific review has always depended on more than the visible article. A computational result may inherit assumptions from source data, calibration, preprocessing, numerical libraries, external services, workflow engines, and domain-specific interpretation. Reproducibility initiatives have made many of these dependencies more inspectable. Open review systems expose reviewer–author exchanges; executable research objects package code, data, and environments; provenance standards describe entities, activities, and agents; and independent execution services record whether identified computations can be run (Smith et al. 2018; Rougier et al. 2017; Nüst and Eglen 2021; World Wide Web Consortium 2013; RO-Crate Community 2025; Leo et al. 2024). These practices address different parts of the research record and should not be collapsed into a single undifferentiated badge.

Agentic systems can help assemble and inspect this record. Published systems already demonstrate multi-step planning, specialized retrieval tools, structured observations, citation-linked synthesis, workflow reconstruction, and automated reproducibility assessment (Grabowski et al. 2026; Riehl et al. 2026). The appropriate conclusion is not that agents can replace scientific reviewers. It is that agents can perform bounded, attributable operations inside a review process whose authority, evidence requirements, and failure states are explicit.

The practical problem addressed here is version binding. A repository link can move. A workflow may be repaired during review. A figure may be rerendered from stored values without recomputing the analysis. A provenance graph may be structurally valid while omitting a manual transformation or mutable service. An author-controlled agent may both produce a result and declare that the result passed. In each case, useful evidence exists, but its scope and authority can be unclear.

MCRP is a prospective protocol intended to make those boundaries inspectable. A research team declares material claims and their evidence paths before a candidate release is accepted. Machine checks and external execution are bound to the exact candidate identity. Qualified humans separately judge whether the data, methods, assumptions, uncertainty treatment, and interpretation warrant the claims. The accepted object is an immutable Scientific Record Release rather than a mutable branch or an unversioned project page. Subsequent changes, dependency drift, or failed re-execution produce explicit new states; they do not silently rewrite the historical record.

This supplement documents a bounded proposal. It defines the review object, the externally enforced lifecycle, assessment modes, resource description, trust boundary, and freshness behavior. It does not claim that claim–evidence graphs are new, that MCRP improves reviewer performance, or that the present reference implementation is scientifically conformant. No comparative study has measured defect detection, claim coverage, reviewer effort, or reviewer agreement under MCRP against simpler review packages; those outcomes remain evaluation targets.

2. Established foundations and the remaining design problem

MCRP builds on established review and execution practices rather than replacing them. JOSS uses transparent, issue-based human review supported by automated repository checks (Smith et al. 2018). ReScience C asks independent reviewers to run an implementation and judge reproduction or replication by domain standards (Rougier et al. 2017). CODECHECK records independent execution of identified computations without claiming to investigate or repair the science (Nüst and Eglen 2021). The ACM artifact policy separately labels availability, artifact quality, and result validation (Association for Computing Machinery 2020). These precedents already establish that

execution evidence and scientific judgment are related but distinct review products.

Claim and evidence graphs are also established. Micropublications represent machine-readable claims, evidence, methods, attribution, support, challenge, disagreement, and transitive evidence closure (Clark et al. 2014). Nanopublications bind an atomic assertion to separate provenance and publication-information graphs, supporting attribution and persistent publication (Kuhn et al. 2018). EVI goes further by extending PROV with defeasible assertions about results, data, methods, and software; a result can have a supporting directed acyclic graph of computations, software, data, and agents, with support and challenge propagated through that graph (Al Manir et al. 2021). MCRP therefore makes no priority claim for claim nodes, evidence edges, support/challenge semantics, immutable atomic records, or graph-shaped computational justification.

Execution and agent provenance are likewise occupied territory. W3C PROV defines entities, activities, agents, derivation, generation, attribution, association, roles, and delegation (World Wide Web Consortium 2013). RO-Crate and Workflow Run RO-Crate package research objects and workflow-run context, including inputs, outputs, software, versions, status, errors, and responsible human agents (RO-Crate Community 2025; Leo et al. 2024). CWLProv captures content-addressed workflow inputs, outputs, and provenance traces (Khan et al. 2019). PROV-AGENT extends PROV to prompts, responses, decisions, human and peer-agent interactions, MCP tools, and downstream workflow outcomes, with near-real-time capture and evaluation across facilities (Souza et al. 2025). MCRP should profile or export these representations rather than create incompatible substitutes.

The closest execution-level collision is Paper-replication, which treats selected paper claims as replication targets and links generated outputs, execution records, provenance, comparison rules, and report coverage. Its persistent workspace and external validation gates determine completion rather than accepting an agent’s assertion. The reported evaluation includes twelve runs and 158 recorded targets (Hans and Billionis 2026). This already realizes much of the broad claim-to-execution-to-report path and external completion checking that could otherwise be attributed to MCRP.

Traxia is the closest collision for broad agent-native publication and review. It proposes signed agent identities, reasoning traces, claim confidence, immutable contribution logs, tiered peer review, human-agent collaboration, reputation, and contradiction detection; its paper reports architectural foundations and a partial prototype rather than empirical validation (Dogah 2026). MCRP consequently makes no priority claim for agent identity, signatures, immutable contribution records, staged review, or contradiction handling. A signature establishes integrity and signer identity, not scientific adequacy or reviewer independence.

The residual MCRP proposal is narrower: a **prospective, externally enforced, claim-versioned lifecycle**. Before execution, a candidate fixes claim-scoped acceptance predicates and required approver roles. A claim transition may then depend on immutable run evidence and a separately authorized check. Approvals are scoped to a release and affected claims; declared dependency changes invalidate the decisions they can reach. Records should export to established claim, provenance, and research-object standards. This lifecycle is an object for evaluation, not a demonstrated improvement and not a claim that its constituent graph or review mechanisms are new. Formal crosswalks and round-trip conformance tests for EVI, Micropublications, Nanopublications, PROV-AGENT, and RO-Crate have not been completed. MCRP also has not been compared directly with Paper-replication or Traxia on prospectively fixed acceptance, mutation-driven invalidation, interruption recovery, independent authorization, false completion, or stale-claim transitions.

3. Operational vocabulary

MCRP separates five properties that are often described with the single word “reproducible.” **Rerenderability** means regenerating a presentation from saved derived values. **Computational repeatability** means rerunning the declared workflow with the same identified inputs and materially equivalent setup. **Computational reproducibility** requires a non-author to reconstruct and execute the identified workflow under reviewer control. **Independent replication** tests the scientific proposition with a materially independent implementation, dataset, measurement, or experimental realization. **Scientific validity** is the qualified judgment that a claim is warranted under its stated scope, assumptions, uncertainties, and domain standards. This vocabulary follows distinctions used by artifact-review and reproduction communities while making the weak rerendering case explicit (Association for Computing Machinery 2020; Rougier et al. 2017).

No one property implies all stronger properties. Bitwise-identical figures may be rerendered from unsupported values. A reproducible computation may implement a biased method. An independent replication may disagree for reasons that require further scientific interpretation. MCRP records orthogonal outcomes such as **schema-valid**, **artifacts-resolved**, **workflow-executed**, **outputs-match**, **claim-covered**, **human-methods-approved**, and **fresh-as-of**. It does not reduce these outcomes to a universal green check.

4. The Scientific Record Release

The unit of review is an immutable **Scientific Record Release** (SRR). An SRR resolves to a claim manifest; artifact and provenance manifests; source, workflow, configuration, and environment descriptions; acceptance tests; trust and access declarations; execution and review records; licenses; resource descriptions; and an archival identity. The candidate may change during review, but acceptance binds to one exact digest and release identifier. A later material change creates a successor SRR.

Each material scientific claim receives a stable identifier within the release. A material claim is one whose removal or revision would change a principal result, interpretation, abstract, conclusion, or safety statement. Its record states the exact text and scope, claim type, supporting or contradicting evidence, claim-bearing values or artifact fragments, producing executions, comparison criteria, uncertainty model, assumptions, limitations, and required reviewer roles. Evidence relations are typed—for example, **supports**, **calibrates**, **bounds**, **contradicts**, or **contextualizes**—so a contextual citation cannot silently become derivational support.

The review graph must be traversable in both directions. A reviewer can move from a claim to evidence, outputs, execution, workflow, configuration, environment, and source inputs. A dependency update or failed artifact can be traced forward to every affected output and claim. Derivation edges must be acyclic, although citation and interpretation links may form cycles when their types are explicit.

Artifacts are classified as raw, intermediate, derived, or presentation objects relative to a declared custody boundary. “Raw” does not mean metaphysically original; it means earliest available within the record. If the workflow begins with provider-calibrated data, that boundary and provider process remain declared trust leaves. Each included object has a content digest, schema or media type, role, size, acquisition or creation time, custodian, access status, and resolver. For an external or restricted object that cannot be hashed directly, the release records the strongest available version identity and the resulting loss of inspectability.

This graph is not asserted to be a new data structure. Its role in MCRP is normative: material claims block acceptance when required paths, identities, tests, or human dispositions are missing. The sufficiency and usability of the proposed required fields have not been evaluated across scientific domains; multi-domain trials are required before treating the present field set as a credible common profile.

5. An externally enforced lifecycle

MCRP’s central position is that verification authority must be distinct from author-controlled production. Separation need not imply a particular institution or service. It means that the verifier’s identity, policy, execution boundary, permissions, and result are recorded independently of the process that produced the candidate claim.

5.1 Candidate construction

Authors assemble a candidate SRR and declare its claim scope. Author-run tests, logs, and attestations are admissible evidence, but they remain author evidence. The candidate includes expected results and tolerances fixed before the verifier’s run, meaningful scientific invariants, and negative or adversarial controls. Silent repairs are prohibited: if a reviewer or agent modifies code, configuration, data, or environment, the patch is preserved and the candidate receives a new identity.

5.2 Admission and machine checks

An external gate resolves the candidate digest and validates schemas, unique identities, graph reachability, artifact digests, observable undeclared inputs and outputs within a named instrumented boundary, acceptance criteria, and freshness policy. Dependencies outside that boundary—including uninstrumented services, manual transformations, and silent upstream processes—remain trust leaves or omissions rather than machine-detectable facts. The gate must fail closed when a required object cannot be resolved. A structurally valid record can still fail later scientific review; admission only establishes that the declared review object is inspectable enough to proceed.

5.3 Reviewer-controlled execution

Where the assessment mode permits execution, a non-author verifier reconstructs the declared environment and runs the claim-bearing workflow or its explicitly bounded surrogate. The execution record captures inputs, source and configuration identities, runtime and dependency state, compute context, external services, timestamps, logs, outputs, and comparison results. Agent assistance is allowed for installation, inventory, execution, comparison, and discrepancy reporting, but every action that changes the candidate is attributable and reviewable.

An author’s agent cannot satisfy this independence condition merely by using a different model or session. Nor can an agent hold scientific sign-off authority. Feasible independence criteria for small collaborations, community services, and institutional review environments remain an unresolved governance question, and the present proposal does not establish that one boundary will work in all three settings.

5.4 Human scientific disposition

Qualified reviewers judge data fitness, method validity, uncertainty, systematics, domain assumptions, counterevidence, and interpretation. Their decision names the reviewed claims, release digest,

competencies, conflicts, unresolved limitations, and residual risks. Machine success and human approval remain separate fields. Acceptance requires all mandatory decisions for the claimed assurance level; it never certifies truth.

5.5 Immutable acceptance and supersession

The accepted SRR is deposited under a version-specific persistent identifier and linked to the review record. A concept DOI or mutable “latest” page may aid discovery but cannot identify the reviewed release (Zenodo, n.d.). Any material revision creates a new candidate and invalidates affected machine and human decisions until they are explicitly re-established. Successor manifests relate claims as **unchanged**, **revised**, **split**, **merged**, **narrowed**, **broadened**, or **retired**; only an **unchanged** claim is eligible for direct carry-forward. Every carry-forward requires a separately authorized scope-delta disposition covering changed claim scope, custody boundaries, plausible undeclared dependencies, and missing graph edges. Declared-graph non-reachability is necessary evidence, not sufficient proof, that a prior decision is unaffected.

5.6 Freshness and failure

Each accepted release declares time-based and event-based freshness triggers: source or data revision, calibration update, dependency or security event, service change, platform retirement, or expiration of a review interval. Where rerunning is feasible, later checks produce new dated evidence without overwriting the original result. A failure preserves both the last passing run and first failing run and assigns a scoped state such as **artifact-unavailable**, **environment-drift**, **data-revision**, **numerical-drift**, **claim-mismatch**, or **unknown**. A decayed environment does not by itself disprove the historical claim; it changes the present reproducibility status.

The practical benefit of this lifecycle remains a hypothesis. No longitudinal study has tested whether explicit supersession and freshness states reduce stale or misleading reproducibility claims.

6. Assurance, resources, and assessment modes

MCRP defines cumulative assurance levels without treating them as truth grades. MCRP-0 is useful disclosure that misses one or more minimum requirements. MCRP-1 is a claim-level, content-identified, provenance-complete-within-the-declared-boundary, tested, archived record with adversarial checks and independent scientific sign-off. MCRP-2 adds non-author reconstruction and execution with independent checking of claim-bearing results. MCRP-3 adds independent method or data challenge, sensitivity to material choices, relevant platform comparisons, and exercised freshness/failure handling.

Assurance is distinct from resource burden. MCRP describes burden with **RRP[C,D,P,X,H,A]**: compute; data and storage; platform; access and governance; human intelligence; and agent intelligence. Each coordinate has an ordinal level and an exact measured or estimated envelope. Profiles are compared componentwise. A workflow requiring more compute but less specialized human expertise may be incomparable with one having the opposite trade. A scalar maximum may be retained as a routing hint, but it is not authoritative and is never an assurance score.

Four assessment modes address practical boundaries. **full** begins at the declared custody boundary and executes the complete claim-bearing path. **slice** applies production algorithms to a justified subset and excludes untested scale or rare regimes. **checkpoint** begins from an identified intermediate and must preserve its generator identity, direct-input lineage, full-run evidence, and equivalence

tests. **witness** records authorized inspection of immutable inputs, scheduler or service records, logs, intermediates, outputs, and checks when independent execution is impossible. Every mode has its own resource profile and per-claim coverage of **full**, **partial**, **audit-only**, or **unassessed**.

Assurance is assigned per claim at a dated assessment, not inferred from the mode alone. MCRP-2 requires **workflow-executed** and **outputs-match** under non-author control for the stated claim scope; **audit-only** or **unassessed** coverage cannot satisfy it, while **slice** or **checkpoint** coverage must remain visibly bounded. MCRP-3 requires the additional challenge and freshness outcomes for that same scope. A release reports the underlying outcome vector, mode, and coverage for every material claim; if it also displays one aggregate level, that label is the minimum current level across those claims. A stale or disputed component removes the current aggregate label while preserving the dated historical result.

These modes are intended to make limitations visible, not to award equivalent status to cheaper substitutes. A checkpoint may validate downstream algorithms while upstream production remains trusted. A witness may provide strong evidence inside a restricted facility without making the data publicly inspectable. No case study has tested whether reviewers interpret the coverage matrix and resource profiles consistently, so the categories should be treated as proposed disclosure fields rather than calibrated review instruments.

7. Reference implementation and current evidence

The current reference implementation is a dependency-light synthetic gravitational-wave-inspired matched-filter workflow. It materializes a deterministic observation, evaluates a small template bank, emits intermediate and derived artifacts, records provenance, and runs off-source, noise-only, calibration-scale, and wrong-template controls. Its manifests exercise claim mapping, evidence identity, run provenance, resource profiles, checkpoint declarations, human-review placeholders, and freshness policy.

The author-reported verification for version 0.1.2 lists four checked claims, nine scientific checks, thirteen passing unit tests, and eleven intentional corruptions rejected for their intended schema, lineage, claim, resource, or authority failures. The present supplement does not contain an itemized public mapping from those counts to named fixtures or checks. The verifier status is deliberately **machine-pass-human-and-archive-pending**. These results support only the statement that the synthetic implementation enforces its encoded invariants on the tested fixtures. They do not establish completeness of the protocol, scientific adequacy of the example, usability, independent reproducibility, or improved review quality. The workflow has not been rerun by an independent party against an immutable public release, its domain invariants have not received qualified scientific review, and no version-specific archive identifier or accountable release attestation exists.

8. Adoption and interoperability

MCRP should be implemented as a profile over existing standards where possible. PROV-compatible relations can represent derivation and responsibility; RO-Crate can package artifacts and context; Workflow Run RO-Crate can represent executions; archival repositories can provide persistent identities; and software supply-chain formats can describe component and build provenance (World Wide Web Consortium 2013; RO-Crate Community 2025; Leo et al. 2024; Zenodo, n.d.). Scientific review requirements remain an additional policy layer because syntactic conformance does not determine evidential adequacy.

A staged adoption path is preferable to mandatory maximal capture. A project can first enumerate principal claims and distinguish presentation from derived values. It can then identify inputs and executions, add independent checks, and finally introduce freshness and higher assurance levels. Legacy records should be annotated conservatively rather than rewritten to imply provenance that was never captured.

Adoption cost is a central unresolved question. Detailed manifests can burden authors and reviewers, and poorly designed automation can create provenance theater rather than useful scrutiny. Domain profiles may improve relevance, but they can also fragment interoperability. Restricted data, proprietary services, and facility-scale computation will remain less inspectable. Incentives for independent verifiers and qualified scientific reviewers are not supplied by a schema. No study has measured MCRP authoring time, researcher experience, reviewer time or comprehension, or maintenance across a dependency or data revision.

9. Limitations and research agenda

MCRP is currently a prospective design supported by a synthetic demonstration. It has not been tested on a submitted or published scientific analysis. No independent group has reconstructed its reference workflow. No qualified reviewer has accepted its human checklist. No archive identity or signed release attestation has been issued. The protocol’s assurance levels and resource axes have not been calibrated across domains, and ordinal levels may conceal important variation even when exact envelopes accompany them.

The claim manifest depends on judgment about materiality and scope. Authors may omit claims, split them strategically, or declare a custody boundary that excludes important upstream transformations. Automated graph completeness checks cannot discover every omitted fact. External enforcement reduces self-certification risk but does not eliminate institutional conflicts, shared dependencies, correlated errors, or verifier mistakes. Human sign-off can be inconsistent, costly, or unavailable. Freshness monitoring can identify declared changes while missing silent upstream drift.

The present proposal also does not address every kind of research. Wet-laboratory work, qualitative research, human-subject studies, and analyses whose critical evidence is tacit or experiential may require different record structures. The protocol must not encourage inappropriate disclosure of personal, proprietary, controlled, or security-sensitive material. Witness modes expose restrictions but do not make restricted evidence public.

An evaluation program should therefore test MCRP on public workflows with different resource profiles; compare it with repository-plus-instructions, provenance-only, and independent-execution baselines; measure detected defects, false findings, reviewer effort, agreement, and recovery after change; and report failures as well as successes. Human-subject review may be required when studying reviewer behavior. Until such work is complete, MCRP should be described as a protocol proposal and reference implementation, not an evidence-backed improvement to peer review.

10. Conclusion

Agentic tools can retrieve evidence, reconstruct environments, execute workflows, and inspect structured records, but their utility depends on an explicit account of what was checked, by whom, against which immutable candidate, and with what remaining human judgment. MCRP proposes a claim-versioned Scientific Record Release and an externally enforced lifecycle that separates author evidence, machine verification, reviewer-controlled execution, and scientific disposition. It also

makes resource limits, restricted modes, trust boundaries, supersession, and freshness visible.

The proposal is deliberately narrower than automated scientific certification. A passing execution is evidence about an execution. A complete declared graph is evidence about the declared record. Human approval is a dated judgment bound to a particular release. None proves a claim true. Any future reconsideration of scholarly publication should therefore follow, not precede, empirical work: an archived protocol version, independent review, interoperability tests, and comparison with simpler review packages on real scientific workflows. Only that evidence could establish whether MCRP is usable or improves review outcomes.

Al Manir, Sadnan, Justin Niestroy, Maxwell Adam Levinson, and Timothy Clark. 2021. “Evidence Graphs: Supporting Transparent and FAIR Computation, with Defeasible Reasoning on Data, Methods, and Results.” In *Provenance and Annotation of Data and Processes*. Springer International Publishing. https://doi.org/10.1007/978-3-030-80960-7_3.

Association for Computing Machinery. 2020. *Artifact Review and Badging, Version 1.1*. <https://www.acm.org/publications/policies/artifact-review-and-badging-current>.

Clark, Tim, Paolo N. Ciccarese, and Carole A. Goble. 2014. “Micropublications: A Semantic Model for Claims, Evidence, Arguments and Annotations in Biomedical Communications.” *Journal of Biomedical Semantics* 5: 28. <https://doi.org/10.1186/2041-1480-5-28>.

Dogah, Wisdom. 2026. *Traxia: A Framework for Verifiable, Agent-Native Scientific Publishing*. <https://arxiv.org/abs/2606.08256v1>.

Grabowski, Piotr, Mohamed Alameen, Jorge Bretones, et al. 2026. *Research Assistant: AstraZeneca’s Agentic System for R&D*. <https://arxiv.org/abs/2608.12395>.

Hans, Atharva, and Ilias Bilionis. 2026. *Coding-Agents Can Replicate Scientific Machine Learning Papers*. <https://arxiv.org/abs/2607.02134v2>.

Khan, Farah Zaib, Stian Soiland-Reyes, Richard O. Sinnott, Andrew Lonie, Carole Goble, and Michael R. Crusoe. 2019. “Sharing Interoperable Workflow Provenance: A Review of Best Practices and Their Practical Application in CWLProv.” *GigaScience* 8 (11). <https://doi.org/10.1093/gigascience/giz095>.

Kuhn, Tobias, Albert Meroño-Peñuela, Alexander Malic, et al. 2018. “Nanopublications: A Growing Resource of Provenance-Centric Scientific Linked Data.” *2018 IEEE 14th International Conference on e-Science (e-Science)*, 83–92. <https://doi.org/10.1109/eScience.2018.00024>.

Leo, Simone, Michael R. Crusoe, Laura Rodríguez-Navas, et al. 2024. “Recording Provenance of Workflow Runs with RO-Crate.” *PLOS ONE* 19 (9): e0309210. <https://doi.org/10.1371/journal.pone.0309210>.

Nüst, Daniel, and Stephen J. Eglen. 2021. “CODECHECK: An Open Science Initiative for the Independent Execution of Computations Underlying Research Articles During Peer Review to Improve Reproducibility.” *F1000Research* 10: 253. <https://doi.org/10.12688/f1000research.51738.2>.

- Riehl, Kevin, Andres L. Marin, Nikofors Zacharof, et al. 2026. *ARA: Agentic Reproducibility Assessment for Scalable Support of Scientific Peer-Review*. <https://arxiv.org/abs/2605.02651>.
- RO-Crate Community. 2025. *RO-Crate Metadata Specification 1.2*. <https://w3id.org/ro/crate/1.2>.
- Rougier, Nicolas P., Konrad Hinsén, Frédéric Alexandre, et al. 2017. “Sustainable Computational Science: The ReScience Initiative.” *PeerJ Computer Science* 3: e142. <https://doi.org/10.7717/peerj-cs.142>.
- Smith, Arfon M., Kyle E. Niemeyer, Daniel S. Katz, et al. 2018. “Journal of Open Source Software (JOSS): Design and First-Year Review.” *PeerJ Computer Science* 4: e147. <https://doi.org/10.7717/peerj-cs.147>.
- Souza, Renan, Amal Gueroudji, Stephen DeWitt, et al. 2025. *PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows*. <https://arxiv.org/abs/2508.02866v3>.
- World Wide Web Consortium. 2013. *PROV-O: The PROV Ontology*. <https://www.w3.org/TR/prov-o/>.
- Zenodo. n.d. *Digital Object Identifier (DOI): Versioning*. <https://help.zenodo.org/docs/deposit/describe-records/reserve-doi/#doi-versioning>.